Rafael Fernandes Lopes

Fullstack Software Engineer · 25+ years experience

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SUMMARY

Fullstack Engineer & Architect with 25+ years of experience delivering scalable web and mobile ecosystems across industry and academia. Specialized in JavaScript/TypeScript (React, Next.js, Node.js) and Python. Proven track record of leading cross-functional teams, launching successful EdTech platforms to thousands of users, and directing advanced R&D projects.

WORK EXPERIENCE — INDUSTRY

Senior Software Engineer

2022 – Present

Fullstack Labs

– Contributed to the design and delivery of production-grade applications across 20+ products for Fueled and Encyclopaedia Britannica, including a goal-planning platform and a digital learning suite serving 140M+ students globally.
– Spearheaded high-ownership features end-to-end — real-time draft-state planning, cross-product SSO authentication, and student progress reporting — in fast-paced, client-facing environments.
– Improved platform performance and reliability — including a refactoring that reduced API calls and cut backend DB round-trips by 50% on high-traffic endpoints, eliminating unnecessary re-renders on admin dashboards, and leading Next.js and Node LTS upgrades across multi-region deployments.

Technologies: React.js, Next.js, Redux, Node.js, Sass, Python, Java.

Founder & Lead Developer

2017 – 2022

ImaginaKIDS — EdTech Startup

– Bootstrapped and scaled platform to 15k+ students, 300+ teachers, and 380k+ pages created; sustained R$490k in annual recurring revenue with sub-10% churn, validating strong product-market fit and hands-on product leadership from zero to revenue.
– Built and scaled web and mobile platforms end-to-end as lead developer.

Technologies: React.js, React Native, Redux, Firebase, Flutter.

Earlier roles (2003–2006): Web Developer at UFMA IT Center and Fapema — built and maintained institutional web systems and information management platforms. *PHP, MySQL, JSP/Servlets, JSF, JPA.*

WORK EXPERIENCE — ACADEMIA

Professor (Adjunct), Researcher & Innovation Leader

2006 – 2022

IFMA (2006–2016) & UFMA (2016–2022)

– Served as Director of the Department of Entrepreneurship & Innovation (DEMI/UFMA), mentoring 10 tech startups with a 70% success rate in securing governmental funding, coordinating 10+ funded R&D projects, and driving institutional entrepreneurship programs — while also serving as peer reviewer for IEEE conferences and Springer's Wireless Personal Communications journal.
– Taught undergraduate and graduate-level Computer Science courses across two major federal institutions for nearly two decades, supervising 7+ Master's students and publishing 6 journal articles and 48+ conference papers spanning distributed systems, IoT, mobile computing, EdTech, and wireless communications.

EDUCATION

Ph.D. in Electrical Engineering · **Information Processing** — **2013**
Federal University of Campina Grande (UFCG)

M.Sc. in Electrical Engineering · **Computer Science** — **2006**
Federal University of Maranhão (UFMA)

B.Sc. in Computer Science — **2003**
Federal University of Maranhão (UFMA)

SKILLS

Expert	React.js, Next.js, React Native, Node.js, Express.js, Firebase, TypeScript, HTML/CSS/JavaScript
Proficient	Flutter, MongoDB, PostgreSQL, Redis, Python, Google Cloud, Amazon AWS, Docker, Jest, Serverless
Applied	Native Modules (Android and iOS), Strapi, WebSocket, Socket.io, gRPC, MQTT, RabbitMQ, FastAPI, Django